

Assignment 1

Question 1:

- a) Explain the difference between a kernel and a user program in an operating system.
- b) Explain how the Operating System acts as a Resource Allocator and a Control Program. Provide one scenario for each where the OS must resolve a conflict.

Question 2:

- a) Compare among CPU Scheduler - Job Scheduler - Swapper
- b) Explain the Zombie state and the Orphan state of a process. Analyze the following two C code snippets. For each, identify if it demonstrates a zombie or orphan process.

Code Snippet A:

```
int main() {
    pid_t pid = fork();
    if (pid > 0) {
        printf("Parent PID: %d\n", getpid());
        exit(0);
    } else if (pid == 0) {
        sleep(10);
        printf("Child PID: %d\n", getpid());
    } else {
        perror("fork");
    }
    return 0;
}
```

Code Snippet B:

```
int main() {
    pid_t pid = fork();
    if (pid > 0) {
        printf("Parent PID: %d\n", getpid());
        sleep(30);
    } else if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        exit(0);
    } else {
        perror("fork");
    }
    return 0;
}
```

Question 3:

- a) Differentiate thread from process.

- b) Imagine designing a web browser application that handles multiple tabs: one tab downloads a large file, another renders a video, and a third processes user input (e.g., scrolling). Explain how you would use multithreading to achieve concurrency on a single-core CPU versus parallelism on a multicore CPU.

Question 4:

The following processes are being scheduled using a preemptive, round robin scheduling algorithm.

Process	Priority	Burst	Arrival
P1	40	20	0
P2	30	25	25
P3	30	25	30
P4	35	15	60
P5	5	10	100
P6	10	10	105

Each process is assigned a numerical priority, with a higher numerical priority, with a higher number indicating a higher relative priority. In addition to the processes listed above, the system also has an **idle task** (which consumes no CPU resources and is identified as *Pidle*). This task has priority 0 and is scheduled whenever the system has no other available processes to run. The length of a time quantum is 10 units. If a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

- Show the scheduling order of the processes using a Gantt chart.
- What is the turnaround time for each process?
- What is the waiting time for each process?

Question 5:

- A financial system has two services: Payment Service and Audit Service. Both services must write records into the same transaction log file. To prevent data corruption, the developer implemented a software mutual-exclusion mechanism based on Peterson's Algorithm. Suppose both services simultaneously request access to the logging module. The Payment Service declares that it wants to write first, but the Audit Service records that the Payment Service should proceed before it. Which service will eventually gain access to the logging module first? Explain how the algorithm ensures mutual exclusion and fairness in this situation.
- Consider a dual-core operating system where two worker processes update a shared in-memory cache. Process A enters the cache-update module after completing the entry protocol described above. Before it exits the critical section, the CPU scheduler switches execution to Process B. Evaluate whether Process B can enter the cache-update module during this time.

